



TECHNOLOGY, SUSTAINABILITY AND PUBLIC POLICY

MILESTONE 2: POLICY MEMO

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1 Executive Summary

The EPFL–UNIL gym faces persistent congestion, with an average hourly inflow of 62 students significantly exceeding the welfare-maximizing arrival rate of 46 students per hour. This imbalance results in non-linear increases in waiting times, reduced workout quality, and a "congestion tax" that erodes collective well-being. To address this, we assess three policy options: (A) mandatory reservations, (B) credit-based rationing, and (C) soft governance through real-time data.

We recommend a phased implementation transitioning from Option C to Option A. Our strategy is grounded in a hybrid ethical framework designed to uphold deontological safety limits, maximize utilitarian aggregate welfare, and enforce egalitarian access for students with rigid schedules. By first introducing real-time occupancy tracking (Option C) to foster user adaptation and data transparency, we establish the foundation for a managed reservation system (Option A). This final system replaces the inefficiency of "open access" with a fair, predictable, and safety-compliant model, ensuring the gym remains a high-utility asset for the entire university community.

2 Background

Our technical report identifies a welfare-maximising arrival rate of approximately 46 students per hour, balancing user satisfaction, resource efficiency, and equitable access. This arrival rate is derived from the welfare-optimal average occupancy of 57 students in the gym, given observed workout durations of 1.23 hours. Current data observations indicate that around 62 students enter the gym per hour, meaning demand exceeds the socially optimal level by a substantial margin. This gap implies persistent overcrowding rather than occasional congestion. As a result, welfare losses increase rapidly, as waiting times rise non-linearly, workout quality deteriorates, and access becomes less fair. A long-term response would be to expand capacity by adding equipment or redesigning space to raise the welfare-optimal occupancy. However, such investments are costly and not feasible in the short run.

The policy challenge is therefore to realign gym usage with welfare-optimal levels through demand-side interventions. This objective aligns with the EPFL Sports Centre's mission to "improve the well-being of each and every one of us" by protecting the quality of the user experience from the "congestion tax" of overcrowding. However, as a public student facility, this realignment must be handled through our hybrid ethical lens. We must move beyond the false dichotomy of "total open access" versus "strict exclusion," seeking instead a system that upholds deontological safety standards and utilitarian efficiency without compromising our egalitarian commitment to inclusive access.

3 Options analysis

Option A: Mandatory reservation with a strict capacity cap

Access to the gym is conditional on a reservation, with real-time check-in/out enforcing a strict occupancy cap aligned with the welfare-optimal level identified in the technical analysis. By directly controlling occupancy, the key decision variable in the model, this option reliably prevents high-utilization regimes and excessive waiting-time losses.

Survey¹ evidence shows that fewer than 10% of users hold the existing off-peak membership, suggesting that pricing-based self-selection has limited behavioural impact. A mandatory reservation system instead actively coordinates demand and makes access more predictable. The main drawback is a shift from in-gym congestion to access constraints. Fairness and capacity utilization can be addressed through a dynamic wait-list that prioritizes users repeatedly unable to secure reservations, alongside no-show penalties to deter precautionary bookings, complemented by limited protected slots for disabled or systematically disadvantaged users.

¹As part of this project, we conducted an on-site survey of 66 gym users at the EPFL–UNIL gym to collect information on usage patterns, and attitudes toward potential policy measures. The survey results were used to complement the technical analysis and to inform the design and assessment of the policy options discussed in this memo. While the sample is not fully representative of the entire gym population, it provides useful insights into user experiences and practical constraints.

Option B: Credit-based access with peak-weighted usage

Members receive a fixed number of access credits per semester, with peak-time visits using up more credits than off-peak visits. By internalising congestion externalities, this option directly targets arrival rates during peak periods while preserving greater flexibility than a strict capacity cap. Compared to Option A, access is not denied outright but becomes progressively more costly during high-demand periods.

However, equity concerns remain for users with limited schedule flexibility, who may be forced to concentrate visits in peak hours and deplete credits faster. Introducing supplementary fees may also affect perceptions of fairness in a student gym. In addition, feedback effects may arise if users who pay additional fees feel entitled to longer sessions, or if revenues reinvested into gym quality increase overall demand.

Option C: Improved status quo through information and experience design

This strategy adopts a "soft governance" approach, prioritizing transparency and user experience over direct regulation. The centerpiece is a real-time occupancy tracking system integrated with entry and exit gates. By requiring students to scan their cards, the facility can provide live crowding data, empowering users to self-select into less crowded periods through non-binding behavioral nudges. To further optimize flow, the policy addresses the technical drivers of congestion identified in our survey, which revealed that 93% of respondents have never received an initiation course (Appendix - Figure 3). This lack of training likely contributes to avoidable delays and longer queues.

To mitigate these inefficiencies, we propose integrating clearer signage and QR-code-based instructional videos on machinery, allowing users to consult safe-use guidance while they wait. Additionally, high-demand zones would be redesigned as "recovery stations" with seating and water to reduce the psychological cost of waiting. While this approach preserves the culture of open access, its effectiveness relies entirely on voluntary user adaptation. Without a mechanism to cap occupancy, it is unlikely to consistently reach welfare-optimal levels. Furthermore, enhancing the gym experience without access constraints may trigger a feedback effect, where improved amenities attract more users and reinforce peak-time congestion.

Recommended action

Our recommended action is a phased implementation transitioning from information-based soft governance (Option C) to a managed reservation system (Option A). This trajectory is guided by a hybrid ethical framework that balances deontological safety, utilitarian welfare, and egalitarian access.

The first phase implements real-time occupancy tracking via student card scans at entry and exit gates. This establishes a foundation of transparency, using non-binding behavioral nudges to encourage voluntary demand shifting. During this stage, we address technical drivers of congestion by improving signage and adding QR codes to machinery for instructional guidance. This phase is critical for empirical data collection and user adaptation, allowing the gym to monitor the gap between current usage and the welfare-optimal capacity without immediately restricting access.

However, because voluntary adaptation alone cannot consistently prevent the peak-time congestion that erodes collective utility, the second phase introduces a mandatory reservation-based access system. This transition allows the institution to strictly uphold its deontological duty to ensure physical safety by enforcing a hard occupancy cap. Simultaneously, it achieves the utilitarian goal of maximizing aggregate welfare by aligning gym density with the optimal levels identified in our technical analysis.

To fulfill our egalitarian commitment to "erasing social barriers," this reservation system rejects financial rationing in favor of transparent, priority-based rules. It includes specific egalitarian adjustments, such as protected slots for students with rigid academic schedules (e.g., labs or student-athletes) and a reserved share for disabled users. To maintain system integrity, QR-code check-ins will monitor attendance, while a redistribution algorithm for no-shows ensures that capacity is never wasted. By adopting this phased approach at the start of the academic year, EPFL can ensure that gym access is not merely open, but is efficient, safe, and fair.

Appendix

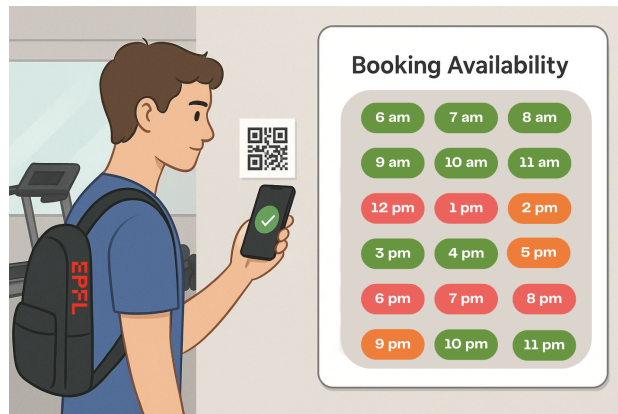


Figure 1: Illustration of gym member scanning QR-code to check-in at reserved gym time slot (Google. n.d.)

Do you have the off-peak membership?

66 réponses

[Copier le graphique](#)

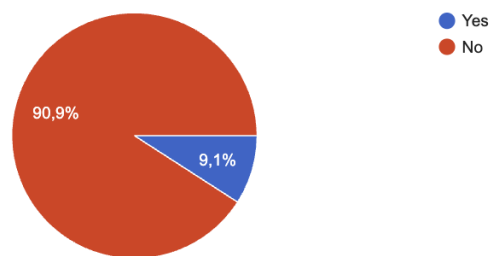


Figure 2: Survey results for off-peak membership demand

Have you received training / an initiation course to use the gym equipment?

66 réponses

[Copier le graphique](#)

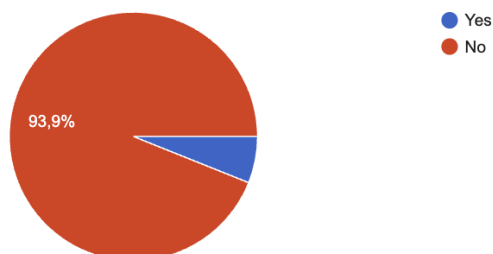


Figure 3: Survey results on Initiation Course to use Equipment

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